

The Clean Slate Architecture Checklist

12 questions to ask before rebuilding any production system. Use this framework on day one of an engagement to establish a scalable, self-healing stack.

01 Zero ClickOps Infrastructure

Are all environments (production, staging, dev) defined 100% as declarative code, preventing any untracked manual modifications via cloud consoles?

TERRAFORM / OPENTOFU

02 Unified Monorepo Context

Are frontend, backend, database migrations, and infrastructure configurations co-located, giving developer tooling and AI agents complete system context?

TURBOREPO / GIT WORKTREE

03 Hermetic Local Sandbox

Can any new engineer spin up a complete, identical local environment in under 15 minutes and with a single terminal command?

DOCKER COMPOSE / DEVCONTAINERS

04 Strict Validation Gate

Is there an automated pre-merge pipeline that blocks code integrates unless lints, type checks, and regression test suites pass cleanly?

GITHUB ACTIONS / CI/CD

05 Continuous Observability

Do we have structured, distributed log tracing and prompt metrics wrapping every core model call and backend API entry point from day one?

LANGFUSE / GRAFANA / PROMETHEUS

06 Zero-Trust Secret Vaulting

Are environment credentials and API tokens stored in a centralized, secure vault rather than distributed inside local .env files or source code?

VAULTWARDEN / DOPPLER / IAM ROLES

07 Declarative Schema Migrations

Are database schema modifications versioned, automated, and designed with backward compatibility to permit zero-downtime rolling upgrades?

PRISMA / LIQUIBASE / FLYWAY

08 Canary & Progressive Rollouts

Can we direct a fractional slice of production traffic (e.g., 2%) to a new version, monitoring health metrics to trigger automatic rollbacks on errors?

ARGOCD / CLOUD RUN TRAFFIC SPLIT

09 Absolute Environment Parity

Are development, staging, and production clusters entirely symmetrical, differing only by computational scale and security access controls?

CONTAINERIZATION / HELM CHARTS

10 Immutable Build Artifacts

Are compilation outputs packaged into cryptographically signed, immutable containers, guaranteeing that what ran in QA is exactly what runs in prod?

OCI CONTAINERS / GOOGLE ARTIFACT REGISTRY

11 Active Configuration Drift Auditing

Is there an automated synchronization process running in the background to detect, alert, and auto-correct any uncodified configuration drift?

GITOPS / TERRAFORM STATE WATCHDOG

12 Agentic Readiness Standard

Is our architecture clear, modular, and declarative enough for autonomous AI coding agents to safely analyze, run, and test code sections in isolation?

SELF-DOCUMENTING CODE / SANDBOX TESTING

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